NANYANG TECHNOLOGICAL UNIVERSITY

SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 1 MH1100 Calculus I Tutorial 5, Week 6

Your tutor will aim to discuss: Problem 1, 4, 5, and 10

Problem 1 Prove the product law: Lef f and g be functions and let $a \in \mathbb{R}$. If the limits

$$\lim_{x \to a} f(x) \qquad \text{and} \qquad \lim_{x \to a} g(x)$$

exist, then

$$\lim_{x \to a} f(x)g(x)$$

exists and

$$\lim_{x \to a} f(x)g(x) = \lim_{x \to a} f(x) \lim_{x \to a} g(x).$$

That is to say, the limit of a product is the product of the limits.

[Solution:]

(1) Preliminary analysis of the problem (guessing a value for δ).

Let $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$. Given any positive number $\epsilon > 0$, we want to find a number δ such that

$$\text{if} \quad 0 < |x - a| < \delta \quad \text{then} \quad |f(x)g(x) - LM| < \epsilon.$$

Now we need to work out an inequality for $|x-a|$ from the second inequality $|f(x)g(x)-LM| < \epsilon$.

But

$$\begin{aligned} |f(x)g(x) - LM| &= |f(x)g(x) + f(x)M - f(x)M - LM| \\ &\le |f(x)g(x) - f(x)M| + |f(x)M - LM| \\ &= |f(x)| \cdot |g(x) - M| + |f(x) - L| \cdot |M| \\ &\le |f(x)| \cdot |g(x) - M| + |f(x) - L| \cdot (|M| + 1) \end{aligned}$$

Since $\lim_{x \to a} f(x) = L$, we can find a positive number δ_1 such that

$$\text{if} \quad 0 < |x - a| < \delta_1 \quad \text{then} \quad |f(x) - L| < \frac{\epsilon}{2(|M| + 1)}.$$

When $0 < |x - a| < \delta_1$, we further have

$$|f(x)| = |f(x) - L + L| \le |f(x) - L| + |L| < \frac{\epsilon}{2(|M| + 1)} + |L|.$$

Similarly, $\lim_{x \to a} g(x) = M$ suggests the existence of a positive number δ_2 such that

$$\text{if} \quad 0 < |x - a| < \delta_2 \quad \text{then} \quad |g(x) - M| < \frac{\epsilon}{2\left(\frac{\epsilon}{2(|M|+1)} + |L|\right)}.$$

We should choose $\delta = \min\{\delta_1, \delta_2\}$.

(2) Proof (showing that this δ works).

Given $\epsilon > 0$, choose $\delta = \min\{\delta_1, \delta_2\}$. If $0 < |x-a| < \delta$, then

$$\begin{aligned}|f(x)g(x) - LM| &\leq |f(x)| \cdot |g(x) - M| + |f(x) - L| \cdot (|M|+1) \\ &< \left[\frac{\epsilon}{2(|M|+1)} + |L|\right] \frac{\epsilon}{2\left(\frac{\epsilon}{2(|M|+1)} + |L|\right)} + \frac{\epsilon}{2(|M|+1)} \cdot (|M|+1) \\ &= \epsilon\end{aligned}$$

Therefore, by the definition of a limit,

$$\lim_{x\to a} f(x)g(x) = \lim_{x\to a} f(x) \lim_{x\to a} g(x).$$

Problem 2 Use the ϵ, δ definition to prove that

$$\lim_{x\to 4} \frac{x^2 - 2x - 8}{x-4} = 6.$$

[Solution:]

(1) Preliminary analysis of the problem (guessing a value for δ).

Let ϵ be a given positive number. We want to find a number δ such that

$$\text{if} \quad 0 < |x-4| < \delta \quad \text{then} \quad |\frac{x^2-2x-8}{x-4} - 6| < \epsilon.$$

Now we need to work out an inequality for $|x-4|$ from the second inequality $|\frac{x^2-2x-8}{x-4} - 6| < \epsilon$.

But if $x \neq 4$

$$\begin{aligned}|\frac{x^2-2x-8}{x-4} - 6| &= |\frac{x^2-2x-8}{x-4} - \frac{6(x-4)}{x-4}| \\ &= |\frac{x^2-2x-8-6(x-4)}{x-4}| \\ &= |\frac{x^2-8x+16}{x-4}| = |x-4|\end{aligned}$$

Therefore we want δ such that

$$\text{if} \quad 0 < |x-4| < \delta \quad \text{then} \quad |x-4| < \epsilon.$$

This suggests that we should choose $\delta = \epsilon$.

(2) Proof (showing that this δ works).

Given $\epsilon > 0$, choose $\delta = \epsilon$. If $0 < |x-4| < \delta = \epsilon$, then

$$|\frac{x^2-2x-8}{x-4} - 6| = |\frac{x^2-2x-8-6(x-4)}{x-4}| = |\frac{x^2-8x+16}{x-4}| = |x-4| < \epsilon$$

Thus

$$\text{if} \quad 0 < |x-4| < \delta \quad \text{then} \quad |\frac{x^2-2x-8}{x-4} - 6| < \epsilon$$

Therefore, by the definition of a limit,

$$\lim_{x\to 4}\frac{x^2-2x-8}{x-4}=6.$$

Problem 3 Use the ϵ, δ definition to prove that

$$\lim_{x\to 2}\left(x^2+2x-7\right)=1.$$

[Solution:]

(1) Preliminary analysis of the problem (guessing a value for δ).

Let ϵ be a given positive number. We want to find a number δ such that

$$\text{if}\quad 0<|x-2|<\delta\quad\text{then}\quad |(x^2+2x-7)-1|<\epsilon.$$

But

$$|(x^2+2x-7)-1|=|x^2+2x-8|=|(x-2)(x+4)|=|x-2|\cdot|x+4|.$$

If $|x-2|<1$ then $|x+4|<7$ and

$$|(x^2+2x-7)-1|=|(x-2)(x+4)|=|x-2|\cdot|x+4|<7|x-2|.$$

If $|x-2|$ is further required to be less than $\epsilon/7$, then

$$|(x^2+2x-7)-1|=|(x-2)(x+4)|=|x-2|\cdot|x+4|<7|x-2|<\epsilon.$$

This suggests that we should choose $\delta=\min\{1,\epsilon/7\}$.

(2) Proof (showing that this δ works).

Given $\epsilon>0$, choose $\delta=\min\{1,\epsilon/7\}$. If $0<|x-2|<\delta$, then

$$|(x^2+2x-7)-1|=|x^2+2x-8|=|(x-2)(x+4)|=|x-2|\cdot|x+4|<\frac{\epsilon}{7}\cdot 7=\epsilon.$$

Thus

$$\text{if}\quad 0<|x-2|<\delta\quad\text{then}\quad |(x^2+2x-7)-1|<\epsilon$$

Therefore, by the definition of a limit,

$$\lim_{x\to 2}(x^2+2x-7)=1.$$

Problem 4 Consider the Heaviside function

$$H(t)=\begin{cases}1, & \text{if } t\geq 0\\ 0, & \text{if } t<0\end{cases}$$

Use the precise definition of a limit to prove that $\lim_{t\to 0}H(t)$ does not exist.

[Solution:] Suppose that $\lim_{t\to 0}H(t)=L$.

- Given ϵ, there exists $\delta>0$ such that if $0<|t-0|<\delta$ then $|H(t)-L|<\epsilon$.

- $|H(t) - L| < \epsilon$ is equivalent to that $L - \epsilon < H(t) < L + \epsilon$.

- For $0 < t < \delta$, $H(t) = 1$, so $1 < L + \epsilon$. That is $L > 1 - \epsilon$.

- For $-\delta < t < 0$, $H(t) = 0$, so $L - \epsilon < 0$. That is $L < \epsilon$. If $0 < \epsilon \leq \frac{1}{2}$, $L < \epsilon \leq \frac{1}{2}$ contradicts $L > 1 - \epsilon \geq \frac{1}{2}$.

Therefore, $\lim_{t \to 0} H(t)$ does not exist.

Problem 5 Let $a > 0$ and n be a positive integer. Prove that

$$\lim_{x \to a} x^{\frac{1}{n}} = a^{\frac{1}{n}}.$$

[Solution:]

(1) Preliminary analysis of the problem (guessing a value for δ).

What we need is some expression in $x^{1/n}$ and $a^{1/n}$ such that:

$$(x^{\frac{1}{n}} - a^{\frac{1}{n}}) \cdot (\text{some expression}) = (x - a).$$

The trick is to exploit the following standard identity, which is true for any real number $p, q \in \mathbb{R}$ and any integer $m > 0$:

$$(p - q)(p^m + p^{m-1}q + p^{m-2}q^2 + \cdots + pq^{m-1} + q^m) = p^{m+1} - q^{m+1}.$$

Note that this identity can be proved by induction. Using this identity we can set $p = x^{1/n}$, $q = a^{1/n}$, and $m = n - 1$. Let

$$\chi = \left(x^{\frac{1}{n}}\right)^{n-1} + \left(x^{\frac{1}{n}}\right)^{n-2}\left(a^{\frac{1}{n}}\right) + \left(x^{\frac{1}{n}}\right)^{n-3}\left(a^{\frac{1}{n}}\right)^2 + \cdots + \left(x^{\frac{1}{n}}\right)\left(a^{\frac{1}{n}}\right)^{n-2} + \left(a^{\frac{1}{n}}\right)^{n-1},$$

it is obvious that when $x > 0$, $\chi > \left(a^{\frac{1}{n}}\right)^{n-1} = a^{\frac{n-1}{n}}$.

Now we have

$$|x^{\frac{1}{n}} - a^{\frac{1}{n}}| = |(x^{\frac{1}{n}} - a^{\frac{1}{n}}) \cdot \frac{\chi}{\chi}| = |\frac{(x^{\frac{1}{n}} - a^{\frac{1}{n}}) \cdot \chi}{\chi}| = |\frac{x - a}{\chi}| < \frac{|x - a|}{a^{\frac{n-1}{n}}}.$$

The above expression suggests that for every given ϵ, if

$$\frac{|x - a|}{a^{\frac{n-1}{n}}} < \epsilon,$$

then

$$|x^{\frac{1}{n}} - a^{\frac{1}{n}}| < \epsilon.$$

And

$$\frac{|x - a|}{a^{\frac{n-1}{n}}} < \epsilon$$

implies that

$$|x - a| < a^{\frac{n-1}{n}}\epsilon.$$

So we can choose $\delta = a^{\frac{n-1}{n}}\epsilon$.

(2) Proof (showing that this δ works).

Given $\epsilon > 0$, choose $\delta = a^{\frac{n-1}{n}}\epsilon$. If $0 < |x-a| < \delta = \delta$, then

$$\begin{aligned} a^{\frac{n-1}{n}}\epsilon = \delta > |x-a| &= |(x^{\frac{1}{n}} - a^{\frac{1}{n}}) \cdot \chi| = |(x^{\frac{1}{n}} - a^{\frac{1}{n}})| \cdot |\chi| \\ &> |x^{\frac{1}{n}} - a^{\frac{1}{n}}| \cdot a^{\frac{n-1}{n}} \end{aligned}$$

Thus

$$\text{if} \quad 0 < |x-a| < \delta \quad \text{then} \quad |x^{\frac{1}{n}} - a^{\frac{1}{n}}| < \epsilon$$

Therefore, by the definition of a limit,

$$\lim_{x\to a} x^{\frac{1}{n}} = a^{\frac{1}{n}}.$$

Problem 6 Let a be a real number. Prove that

$$\text{(a)} \lim_{x\to a^-} \frac{1}{(x-a)^3} = -\infty \qquad\qquad \text{(b)} \lim_{x\to a^+} \frac{1}{(x-a)^3} = \infty.$$

[Solution:]

(a) Let N be a given negative number. We want to find a number $\delta > 0$ such that

$$\text{if} \quad -\delta < x-a < 0 \quad \text{then} \quad \frac{1}{(x-a)^3} < N.$$

But

$$\frac{1}{(x-a)^3} < N \Longleftrightarrow (x-a)^3 > \frac{1}{N} \Longleftrightarrow x-a > \sqrt[3]{\frac{1}{N}}$$

So if we choose $\delta = -\sqrt[3]{\frac{1}{N}}$ and $-\delta < x-a < 0$, then $1/(x-a)^3 < N$.

This shows that

$$\lim_{x\to a^-} \frac{1}{(x-a)^3} = -\infty.$$

(b) Let M be a given positive number. We want to find a number $\delta > 0$ such that

$$\text{if} \quad 0 < x-a < \delta \quad \text{then} \quad \frac{1}{(x-a)^3} > M.$$

But

$$\frac{1}{(x-a)^3} > M \Longleftrightarrow (x-a)^3 < \frac{1}{M} \Longleftrightarrow \sqrt[3]{(x-a)^3} < \sqrt[3]{\frac{1}{M}} \Longleftrightarrow x-a < \sqrt[3]{\frac{1}{M}}$$

So if we choose $\delta = \sqrt[3]{\frac{1}{M}}$ and $0 < x-a < \delta = 1/\sqrt[3]{M}$, then $1/(x-a)^3 > M$.

This shows that

$$\lim_{x\to a^+} \frac{1}{(x-a)^3} = \infty.$$

Problem 7 Use the ϵ, δ definition to prove that

$$\lim_{x\to 0}|x| = 0.$$

[Solution:]

(1) Preliminary analysis of the problem (guessing a value for δ).

Let ϵ be a given positive number. We want to find a number δ such that

$$\text{if} \quad 0 < |x-0| < \delta \quad \text{then} \quad ||x|-0| < \epsilon.$$

Now we need to work out an inequality for $|x-0| = |x|$ from the second inequality $||x|-0| < \epsilon$.

But

$$||x|-0| = ||x|| = |x|$$

Therefore we want δ such that

$$\text{if} \quad 0 < |x| < \delta \quad \text{then} \quad |x| < \epsilon.$$

This suggests that we should choose $\delta = \epsilon$.

(2) Proof (showing that this δ works).

Given $\epsilon > 0$, choose $\delta = \epsilon$. If $0 < |x-0| < \delta = \epsilon$, then

$$||x|-0| = ||x|| = |x| < \epsilon$$

Thus

$$\text{if} \quad 0 < |x-0| < \delta \quad \text{then} \quad ||x|-0| < \epsilon$$

Therefore, by the definition of a limit,

$$\lim_{x\to 0}|x| = 0.$$

Problem 8 Use the definition of continuity to show that $f(x) = |x|$ is continuous at any point $x = a$ of its domain.

[Solution:] To begin, observe that the domain of f is $\mathbb{R}$. We need to show that for every point $a \in \mathbb{R}$, $\lim_{x\to a} f(x) = f(a)$. In detail,

(1) When $a > 0$, $\lim_{x\to a}|x| = \lim_{x\to a} x = a$ exists and equals $|a|$.

(2) When $a < 0$, $\lim_{x\to a}|x| = \lim_{x\to a}(-x) = -a$ exists and equals $|a|$.

(3) When a=0, as shown in the previous question, $\lim_{x\to 0}|x| = 0$.

So we conclude that $f(x) = |x|$ is continuous at any point $x = a$ of its domain.

Problem 9 Use the definition of continuity and standard properties of limits to show that the function is continuous at the given number a

$$(a)\quad f(x) = x^2 + \sqrt{7-x}, \quad a = 4; \qquad (b)\quad g(t) = \frac{t^2+5t}{2t+1}, \quad a = 2;$$

$$(c)\quad p(v) = 2\sqrt{3v^2+1}, \quad a = 1; \qquad (d)\quad f(x) = 3x^4 - 5x + \sqrt[3]{x^2+4}, \quad a = 2.$$

[Solution:]

(a) To begin, observe that the domain of f is $\{x \leq 7\}$. We need to show that for the given number $a = 4$, $\lim_{x\to 4} f(x) = f(4)$. In detail,

$$\begin{aligned}\lim_{x\to 4} f(x) &= \lim_{x\to 4}(x^2 + \sqrt{7-x}) = \lim_{x\to 4} x^2 + \lim_{x\to 4}\sqrt{7-x} \\ &= \left(\lim_{x\to 4} x\right)^2 + \sqrt{\lim_{x\to 4}(7-x)} = \left(\lim_{x\to 4} x\right)^2 + \sqrt{\lim_{x\to 4} 7 - \lim_{x\to 4} x} \\ &= 4^2 + \sqrt{7-4} = 16 + \sqrt{3}.\end{aligned}$$

We also have $f(4) = 16 + \sqrt{3}$. So, $\lim_{x\to 4} f(x) = f(4)$ and $f(x)$ is continuous at $a = 4$.

(b) The domain of g is $\{t \in \mathbb{R} | t \neq \frac{1}{2}\}$. We need to show that for the given number $a = 2$, $\lim_{t\to 2} g(t) = g(2)$. In detail,

$$\begin{aligned}\lim_{t\to 2} g(t) &= \lim_{x\to 2}\frac{t^2+5t}{2t+1} = \frac{\lim_{t\to 2}(t^2+5t)}{\lim_{t\to 2}(2t+1)} = \frac{\lim_{t\to 2} t^2 + \lim_{t\to 2}(5t)}{\lim_{t\to 2} 2t + \lim_{t\to 2} 1} \\ &= \frac{\left(\lim_{t\to 2} t\right)^2 + 5\lim_{t\to 2} t}{2\lim_{t\to 2} t + \lim_{t\to 2} 1} = \frac{2^2 + 5\times 2}{2\times 2 + 1} \\ &= \frac{14}{5}.\end{aligned}$$

We have $g(2) = \frac{14}{5}$. So, $\lim_{t\to 2} g(t) = g(2)$ and therefore $g(t)$ is continuous at $a = 2$.

(c) The domain of p is $\{v \in \mathbb{R}\}$. We need to show that for the given number $a = 1$, $\lim_{v\to 1} p(v) = p(1)$. In detail,

$$\begin{aligned}\lim_{v\to 1} p(v) &= \lim_{v\to 1} 2\sqrt{3v^2+1} = 2\lim_{v\to 1}\sqrt{3v^2+1} = 2\sqrt{\lim_{v\to 1}(3v^2+1)} \\ &= 2\sqrt{\lim_{v\to 1}(3v^2) + \lim_{v\to 1} 1} = 2\sqrt{3\lim_{v\to 1}(v^2) + \lim_{v\to 1} 1} \\ &= 2\sqrt{3\left(\lim_{v\to 1} v\right)^2 + \lim_{v\to 1} 1} = 2\sqrt{3\times(1)^2+1} = 4.\end{aligned}$$

We have $p(1) = 4$. So, $\lim_{v\to 1} p(v) = p(1)$ and therefore $p(v)$ is continuous at $a = 1$.

(d) The domain of f is $\{x \in \mathbb{R}\}$. We need to show that for the given number $a = 2$, $\lim_{x\to 2} f(x) = f(2)$. In detail,

$$\begin{aligned}\lim_{x\to 2} f(x) &= \lim_{x\to 2}(3x^4 - 5x + \sqrt[3]{x^2+4}) = \lim_{x\to 2}(3x^4 - 5x) + \lim_{x\to 2}\sqrt[3]{x^2+4} \\ &= \lim_{x\to 2} 3x^4 - \lim_{x\to 2} 5x + \sqrt[3]{\lim_{x\to 2}(x^2+4)} = 3\lim_{x\to 2} x^4 - 5\lim_{x\to 2} x + \sqrt[3]{\lim_{x\to 2} x^2 + \lim_{x\to 2} 4} \\ &= 3\left(\lim_{x\to 2} x\right)^4 - 5\lim_{x\to 2} x + \sqrt[3]{\left(\lim_{x\to 2} x\right)^2 + \lim_{x\to 2} 4} \\ &= 3 \times 2^4 - 5 \times 2 + \sqrt[3]{2^2+4} = 40.\end{aligned}$$

We have $f(2) = 40$. So, $\lim_{x\to 2} f(x) = f(2)$ and therefore $f(x)$ is continuous at $a = 2$.

Problem 10 Explain why the function is discontinuous at the given number a. Sketch the graph of the function.

$(a)\quad f(x) = \dfrac{1}{x+2}, \quad a = -2;$

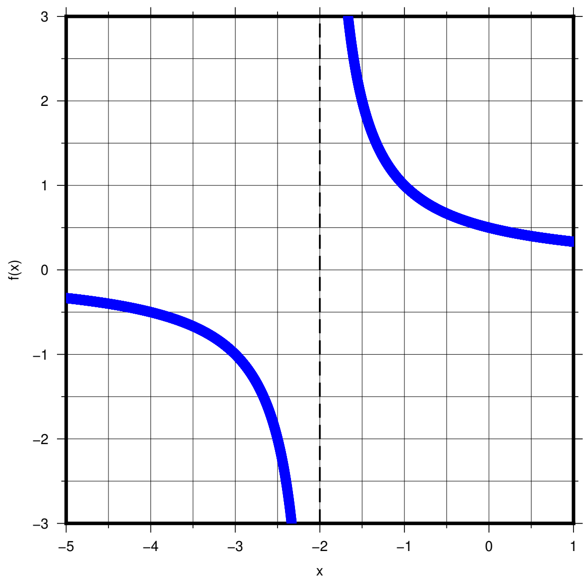
$(b)\quad f(x) = \begin{cases} \frac{x^2-x}{x^2-1} & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}, \quad a = 1;$

$(c)\quad f(x) = \begin{cases} \cos x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 - x^2 & \text{if } x > 0 \end{cases}, \quad a = 0.$

$(d)\quad f(x) = \begin{cases} \frac{2x^2-5x-3}{x-3} & \text{if } x \neq 3 \\ 6 & \text{if } x = 3 \end{cases}, \quad a = 3.$

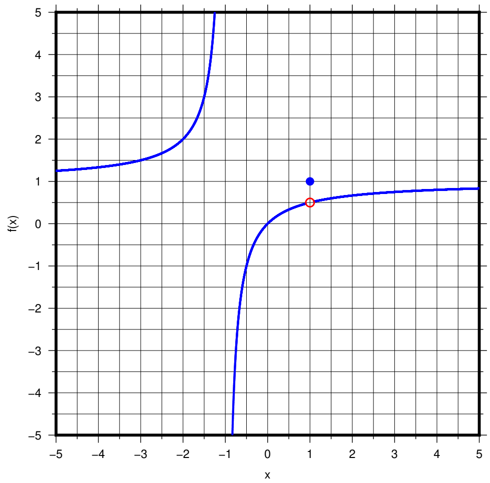
[Solution:]

(a) $f(x)$ is not continuous at $x = -2$ because it is not defined at $x = -2$.



(b) $f(x)$ is not continuous at $x = 1$ because $\lim_{x\to 1} f(x) \neq f(1)$. The details are

$$\lim_{x\to 1} f(x) = \lim_{x\to 1}\frac{x^2-x}{x^2-1} = \lim_{x\to 1}\frac{x(x-1)}{(x+1)(x-1)} = \lim_{x\to 1}\frac{x}{x+1} = \frac{1}{2} \neq 1 = f(1).$$



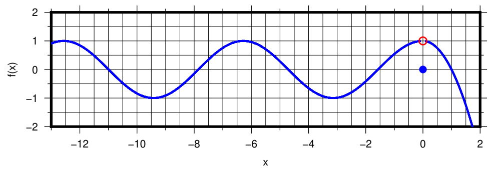
(c) $f(x)$ is not continuous at $x = 0$ because $\lim_{x \to 0} f(x) \neq f(0)$. We have

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \cos x = 1$$

and

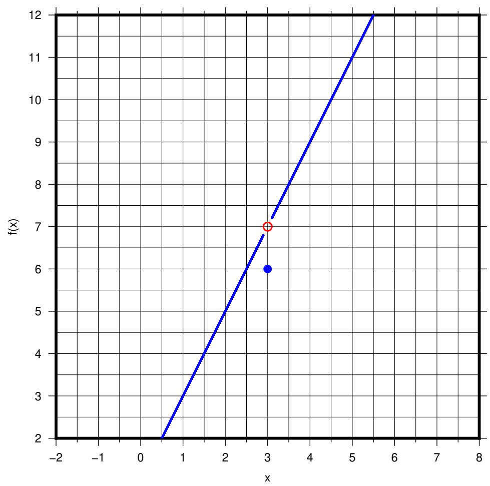
$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (1 - x^2) = 1.$$

So, $\lim_{x \to 0} f(x) = 1 \neq 0 = f(0)$.



(d) $f(x)$ is not continuous at $x = 3$ because $\lim_{x \to 3} f(x) \neq f(3)$. The details are

$$\lim_{x \to 3} f(x) = \lim_{x \to 3} \frac{2x^2 - 5x - 3}{x - 3} = \lim_{x \to 3} \frac{(2x+1)(x-3)}{x-3} = \lim_{x \to 3} (2x + 1) = 7 \neq 6 = f(3).$$



Problem 11 How would you 'remove the discontinuity' of f? In other words, how would you define $f(2)$ in order to make f continuous at 2.

$$(a) \quad f(x) = \frac{x^2 + 2x - 8}{x - 2} \qquad\qquad (b) \quad f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}.$$

[Solution:]

(a) The domain of f is $\{x \in \mathbb{R} | x \neq 2\}$. $f(x)$ is not defined at $x = 2$. We evaluate the limit of $f(x)$ at $x = 2$.

$$\lim_{x \to 2} f(x) = \lim_{x \to 2} \frac{x^2 + 2x - 8}{x - 2} = \lim_{x \to 2} \frac{(x-2)(x+4)}{x - 2} = \lim_{x \to 2} (x + 4) = 6.$$

$f(x)$ has a removable discontinuity at $x = 2$. If we define $f(2) = 6$, this discontinuity can be removed.

(b) The domain of f is $\{x \in \mathbb{R} | x \neq 2, x \neq -3\}$. $f(x)$ is not defined at $x = 2$ and $x = -3$. We evaluate the limit of $f(x)$ at $x = 2$.

$$\lim_{x \to 2} f(x) = \lim_{x \to 2} \frac{x^2 - 3x + 2}{x^2 + x - 6} = \lim_{x \to 2} \frac{(x-2)(x-1)}{(x-2)(x+3)} = \lim_{x \to 2} \frac{x - 1}{x + 3} = \frac{1}{5}.$$

$f(x)$ has a removable discontinuity at $x = 2$. If we define $f(2) = \frac{1}{5}$, this discontinuity can be removed.